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Dual flap pressure relief valve

Abstract

A pressure vent valve for ventilation of the interior of a motor vehicle, with a frame with at least one sealing surface and with at least one flexible valve flap mounted on the frame. In an idle position, the valve flap assembly abuts the sealing surface to prevent a flow of air through the pressure vent valve. When a sufficient pressure occurs, the valve flap assembly assumes an open position lifted from the sealing surface to allow a flow of air through the pressure vent valve. The valve flap assembly comprises a first flap sheet stacked upon a second flap sheet.

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Background/Summary

CROSS-REFERENCE (1) The present application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application Ser. No. 63/256,758, which was filed on Oct. 18, 2021 and is entitled “Dual Flap PRV.” The contents of which are hereby incorporated by reference.

BACKGROUND
(1) The disclosure relates to a return air blocking device for ventilation of the interior of a vehicle, such as a motor vehicle. Such return air blocking devices are fitted in openings in the body of a vehicle. In operation, if a positive pressure predominates in the interior of the vehicle relative to the environment of the vehicle, the valve flaps of the return air blocking device open to allow air to flow out of the vehicle interior and reduce the positive pressure. In the opposite flow direction, the valve flaps block the openings. Example return air blocking devices to ventilate the interior of

vehicles are described in commonly owned U.S. Pat. No. 8,328,609 to Daniel Schneider and U.S. Pat. No. 10,391,838 to Nikolaus Schwarzkopf and Marco Spanier.

(2) Existing return air blocking devices, however, often employ flaps or flap assemblies that can become warped or damaged over time due to dust and moisture cycling, thus resulting in a poor seal. Therefore, it would be desirable to provide a return air blocking device with flap assemblies that mitigate the effects of such warping or damage to maintain a seal over time.

SUMMARY

(3) The present disclosure relates generally to a return air blocking device with flaps that offer increased resistance to warping or damage, substantially as illustrated by and described in connection with at least one of the figures, as set forth more completely in the claims.

Description

DRAWINGS

(1) The foregoing and other objects, features, and advantages of the devices, systems, and methods described herein will be apparent from the following description of particular examples thereof, as illustrated in the accompanying figures; where like or similar reference numbers refer to like or similar structures. The figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the devices, systems, and methods described herein.

(2) FIGS. **1***a* and **1***b* illustrate, respectively, assembly and assembled views of an example return air blocking system in accordance with an aspect of the subject disclosure.

(3) FIGS. **2***a* and **2***b* illustrate, respectively, front and rear perspective views of an example pressure vent valve in accordance with an aspect of the subject disclosure.

(4) FIGS. **2***c*, **2***d*, and **2***e* illustrate, respectively, side elevation, top plan, and front elevational views of the example pressure vent valve.

(5) FIGS. **3***a* and **3***b* illustrate cross-sectional views of the example pressure vent valve taken along cutlines A-A and B-B, respectively.

(6) FIG. **4***a* illustrates a perspective view of an example valve flap assembly in an assembled state.

(7) FIG. **4***b* illustrates a cross-sectional view of the valve flap assembly taken along cutline C-C.

(8) FIG. **4***c* illustrates a perspective view of an example valve flap assembly in a disassembled state.

(9) FIG. **4***d* illustrates a cross-sectional view of the valve flap assembly taken along cutline D-D.

(10) FIG. **5***a* illustrates a perspective view of an example valve flap assembly in accordance with another aspect of the subject disclosure.

(11) FIG. **5***b* illustrates a front elevational view of the example valve flap assembly of FIG. **5***a* in an assembled state.

(12) FIG. **5***c* illustrates a side elevational view of the example valve flap assembly of FIG. **5***a* in the assembled state.

DESCRIPTION

(13) References to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context. Recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all values falling within and/or including the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. In the following description, it is understood that terms such as “first,” “second,” “top,” “bottom,” “side,” “front,” “back,” “upper,” “lower,” and the like are words of convenience and are not to be construed as limiting terms. For example, while in some examples a first side is

located adjacent or near a second side, the terms “first side” and “second side” do not imply any specific order in which the sides are ordered.

(14) The terms “about,” “approximately,” “substantially,” or the like, when accompanying a numerical value, are to be construed as indicating a deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Ranges of values and/or numeric values are provided herein as examples only, and do not constitute a limitation on the scope of the disclosure. The use of any and all examples, or exemplary language (“e.g.,” “such as,” or the like) provided herein, is intended merely to better illuminate the disclosed examples and does not pose a limitation on the scope of the disclosure. The terms “e.g.,” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed examples.

(15) The term “and/or” means any one or more of the items in the list joined by “and/or.” As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y, and/or z” means “one or more of x, y, and z.”

(16) Disclosed is a return air blocking device with flap assemblies that mitigate the effects of flap warping or damage. In one example, a pressure vent valve for ventilation of an interior of a vehicle comprises a frame defining a vent opening, at least one sealing surface, and a plurality of fastening projections, wherein the at least one sealing surface and the plurality of fastening projections are positioned adjacent the vent opening; and a valve flap assembly mounted on the frame at a bearing surface via the plurality of fastening projections. The valve flap assembly comprises a first flap sheet stacked upon a second flap sheet. In an idle position, the valve flap assembly abuts the sealing surface and is configured to block a flow of air through the vent opening. When a pressure increases in the interior of a vehicle, the valve flap assembly assumes an open position where the valve flap assembly is lifted from the sealing surface to allow the flow of air through the vent opening.

(17) In another example, a pressure vent valve for ventilation of an interior of a vehicle comprises a frame defining a plurality of vent openings and a plurality of valve flap assemblies mounted on the frame. Each of the plurality of vent openings comprises a sealing surface positioned adjacent the vent opening. Each of the plurality of valve flap assemblies comprising a first flap sheet stacked upon a second flap sheet. At least one of the plurality of valve flap assemblies is coupled to the frame at each of the plurality of vent openings and configured to alternate between an idle position and an open position. In the idle position, the valve flap assembly is configured to block a flow of air through the vent opening. In the open position, the valve flap assembly is lifted from the sealing surface to allow the flow of air through the vent opening.

(18) In some examples, the valve flap assembly has a thickness between about 0.75 mm and 1.25 mm or about 1.0 mm. In some examples, the pressure vent valve further comprises a second valve flap assembly mounted on the frame and configure to block a flow of air through a second vent opening formed in the frame. In some examples, the frame is fabricated from a rigid material and the valve flap assembly is fabricated from a flexible material. In some examples, the flexible material is a closed pore, foamed material. In some examples, the flexible material is a plastic material or a rubber material. In some examples, the frame further defines a stop surface and the valve flap assembly is secured to the frame such that the valve flap assembly assumes the open position by bending at a location that is spaced from the plurality of fastening projections. In some examples, the stop surface extends over a portion of the valve flap assembly such that, upon assuming the open position, the portion covered by the stop surface is undeflected. In some examples, the valve flap assembly bends in a curved manner when in the open position.

(19) FIGS. **1***a* and **1***b* illustrate, respectively, assembly and assembled isometric views of an

example return air blocking system **100** in accordance with an aspect of this disclosure. The illustrated return air blocking system **100** generally comprises a bracket **102** and a pressure vent valve **104**. The bracket **102** is configured to retain and secure the pressure vent valve **104** relative to a vehicle. The return air blocking system **100** is positioned between an interior of a vehicle and the exterior environment of the vehicle and used for ventilation of the interior. The bracket **102** maybe attached to, or integral with, the vehicle. The vehicle may be, for example, a motor vehicle. In operation, the pressure vent valve **104** (e.g., via the frame **106**) is inserted in an opening provided therefor in the body of the vehicle. The pressure vent valve **104** is inserted such that the back of the pressure vent valve **104** faces the interior of the vehicle, and the front of the pressure vent valve **104** is allocated to a region outside the vehicle interior.

(20) FIGS. **2***a* through **2***d* illustrate, respectively, front perspective, rear perspective, side elevation, and top plan views of the pressure vent valve **104**. FIG. **2***e* illustrates a front elevational view of the frame **106** with the valve flap assemblies **108** omitted for illustrative purposes. The pressure vent valve **104** includes a frame **106** (or similar housing) that defines a generally rectangular form (though other shapes are contemplated). In the illustrated example, the frame **106** defines, inter alia, at least one vent opening **206**, at least one sealing surface **208**, and a plurality of fastening projections **210**.

(21) Depending on the application, the frame **106** may be fabricated from a rigid material, such as, for example, synthetic or semi-synthetic polymers (e.g., plastics, such as acrylonitrile butadiene styrene (ABS) and polyvinyl chloride (PVC), etc.), composite materials (e.g., fiber glass), metal (or a metal alloy), or a combination thereof. In the illustrated example, the frame **106** can be fabricated using a relatively hard plastic via a plastic injection molding process. Therefore, in some examples, the frame **106** and its various features can be formed as a unitary structure.

(22) As best illustrated in FIG. **2***e*, the frame **106** can be divided vertically by one or more longitudinal webs **204***a* and/or horizontally by one or more lateral webs **204***b* to define the one or more vent openings **206**. In the illustrated example, the four vent openings **206** are arrange in a quad arrangement. Other arrangements, however, are contemplated (e.g., in a row, triangular, etc.). In particular, the illustrated frame **106** provides a longitudinal web **204***a* and a lateral web **204***b* to define forms four vent openings **206**. The vent openings **206** are illustrated as generally rectangular openings, though other shapes are contemplated (e.g., square, trapezoidal, etc.).

(23) The frame **106** includes at least one sealing surface **208** and at least one flexible valve flap assembly **108** mounted on the frame **106**. Each of the plurality of valve flap assemblies **108** is coupled to the frame **106** at or adjacent each of the plurality of vent openings **206**. As illustrated, a valve flap assembly **108** is arranged adjacent each of the four vent openings **206** and configured to cover each respective one of the four vent openings **206** when in the idle position. Each of the valve flap assemblies **108** is configured to alternate between the idle position and an open position.

(24) The frame **106** is formed with sealing surfaces **208** such that the valve flap assemblies **108**, when in the idle position, substantially and/or completely prevent the flow of air through the vent openings **206** of the frame **106**. Therefore, the sealing surfaces **208** are generally located at the contact regions between the valve flap assembly **108** and frame **106**. The sealing surfaces **208** for the valve flap assemblies **108** can be fabricated using a softer plastic material. For example, the sealing surfaces **208** can be molded onto the frame **106** in a multi-component injection molding process. In the illustrated example, the one or more sealing surfaces **208** are provided at the perimeter of each of the vent openings **206**. The sealing surface **208** around each vent opening **206** may be continuous or segmented. Therefore, in some examples, the sealing surface **208** generally surrounds the vent openings **206** (or at least a bottom edge) and contacts the valve flap assembly **108**.

(25) The valve flap assemblies **108** are fabricated using a material that is flexible and/or softer relative to the material of the frame **106**. In some examples, the flexible valve flap assembly **108** is made of a flexible material, such as closed pore, foamed material. It is also conceivable that the

valve flap assemblies **108** could comprise other plastic and/or rubber flexible materials. In any case, the valve flap assemblies **108** are generally thin and hence very flexible to allow them to alternate between the idle position and the open position.
(26) FIGS. **3***a* and **3***b*, which are cross-sectional views taken along cutlines A-A (FIG. **2***d*) and B-B (FIG. **2***d*), illustrate the valve flap assemblies **108** in the idle position. The valve flap assemblies **108** are mounted at one end (e.g., along one edge) on the frame **106** via, inter alia, one or more fastening projections **210** formed on the frame **106**. In the illustrated example, a plurality of fastening projections **210** are arranged and aligned in a row along an upper edge of each of the vent openings **206**.
(27) As best illustrated in FIG. **3***a*, an attachment edge **212***a* of each valve flap assembly **108** is coupled to the frame **106** at a bearing surface **302** via a bracket **304**. The attachment edge **212***a* of the valve flap assembly **108** is, in effect, sandwiched between the bearing surface **302** and an inner surface of the bracket **304**. To prohibit the attachment edge **212***a* of the valve flap assembly **108** from sliding out of the bracket **304**, the plurality of fastening projections **210** can pass through the valve flap assembly **108** and into the bracket **304**.
(28) As illustrated, the valve flap assemblies **108** each have a plurality of flap openings **308** formed along the attachment edge **212***a* through which the fastening projections **210** of the frame **106** are passed to secure the valve flap assembly **108** to the frame **106**. The valve flap assemblies **108** are mounted by passing the fastening projections **210** through the flap openings **308**. The fastening projections **210** may be tapered and sized such that a tip of the fastening projection **210** is smaller than the flap opening **308**, while a base end of the fastening projection **210** is larger than the flap openings **308**. As a result, when installed, the flap opening **308** can stretch and/or deform to better contact and secure to the fastening projections **210** at the base end.
(29) The various figures illustrate the valve flap assembly **108** in the idle position (e.g., closed). In the idle position, a free edge **212***b* of the valve flap assembly **108** abuts the sealing surface **208** and is configured to block a flow of air through the vent opening **206**. When a pressure increases in the interior of a vehicle, the valve flap assembly **108** assumes the open position such that the free edge **212***b* of the valve flap assembly **108** is lifted from the sealing surface **208** to allow a flow of air through the vent opening **206**.
(30) When the air pressure increases within the cabin, for example, the valve flap assemblies **108** assume a vent position whereby the free edges **212***b* of the valve flap assemblies **108** flex upward as indicated by arrow **310** to allow air flow through the vent opening **206** of the frame **106**. In other words, when a sufficient pressure occurs, assumes an open position lifted from the sealing surface **208** in which it allows a flow of air through the pressure vent valve **104**. Once the pressure between the vehicle interior and the environment has balanced, the valve flap assemblies **108** close automatically under their own weight (i.e., they return to the idle position).
(31) The pressure vent valve **104** may further comprise one or more stop surfaces **202** (illustrated as tabs), which may be part of the frame **106** and/or bracket **304**. In one example, the valve flap assembly **108** can be secured to the frame **106** such that the valve flap assembly **108** assumes the open position by bending at a location that is spaced from the plurality of fastening projections **210**. As a result, wear and tear of the valve flap assembly **108** at the flap openings **308** is mitigated by shifting the bending/flexing moments away from the attachment point (i.e., flap openings **308**). In the illustrated examples, the stop surface **202** extend over a portion **312** of the valve flap assembly **108** such that, upon assuming the open position, the portion **312** covered by the stop surface **202** is undeflected (e.g., generally flat). As a result, the valve flap assembly **108** can bend in a generally curved manner when in the open position.
(32) FIG. **4***a* illustrates a perspective view of an example valve flap assembly **108** in an assembled state, while FIG. **4***b* illustrates a cross-sectional view of the valve flap assembly **108** taken along cutline C-C. As illustrated, the valve flap assembly **108** is a multi-layer assemble that comprises multiple layers. For example, as illustrated, the valve flap assembly **108** comprises a first flap sheet

108*a* stacked (or layered) upon a second flap sheet **108***b*. The first flap sheet **108***a* and the second flap sheet **108***b* and generally identical to one another and individually joined to the frame **106** via the fastening projections **210** and the plurality of flap openings **308** formed along the attachment edge **212***a* of the valve flap assembly **108**. The first flap sheet **108***a* and the second flap sheet **108***b* may be formed using, for example, a stamping process.

(33) FIG. **4***c* illustrates a perspective view of an example valve flap assembly **108** in a disassembled state, while FIG. **4***d* illustrates a cross-sectional view of the valve flap assembly **108** taken along cutline D-D. As illustrated, the first flap sheet **108***a* and the second flap sheet **108***b* are separate from one another. That is, they are not glued or otherwise adhered to one another to form a unitary structure—as would be the case with a laminated structure. As a result, if one of the first flap sheet **108***a* or the second flap sheet **108***b* becomes damaged and/or curls away from the sealing surface **208**, the other one of the first flap sheet **108***a* or the second flap sheet **108***b* would remain to seal the vent openings **206**. In effect, providing both the first flap sheet **108***a* and the second flap sheet **108***b* allows one to serve as a sacrificial layer—typically the outermost sheet (illustrated as the first flap sheet **108***a*).

(34) In some examples, the valve flap assembly **108** has a thickness between about 0.75 mm and 1.25 mm, or about 1.0 mm. The first flap sheet **108***a* and the second flap sheet **108***b* of a given valve flap assembly **108** may be identical to one another in terms of material, shape, and dimensions. Therefore, each of the first flap sheet **108***a* and the second flap sheet **108***b* can have a thickness between about 0.375 mm and 0.75 mm, or about 0.5 mm. In some examples, 3 or more flap sheets may be used; however, attention must be paid to the overall thickness and/or rigidity of the valve flap assembly **108** to ensure that the valve flap assembly **108** remains sufficiency flexible to deflect into the open position during operation.

(35) FIG. **5***a* illustrates a perspective view of an example valve flap assembly **108** in accordance with another aspect of the subject disclosure. As illustrated, the valve flap assembly **108** is in a disassembled state. FIGS. **5***b* and **5***c* illustrate, respectively, a front elevational view and a side elevational view of the valve flap assembly **108** in an assembled state. While the above-described examples illustrate and describe where the first flap sheet **108***a* and the second flap sheet **108***b* of a given valve flap assembly **108** are identical to one another in terms of material, shape, and dimensions, the first flap sheet **108***a* and the second flap sheet **108***b* may instead be different in terms of material, shape, and dimensions.

(36) In the illustrated example, the first flap sheet **108***a* is larger than the second flap sheet **108***b*. Specifically, the first flap sheet **108***a* is illustrated as having a first width (W.sub.1) and a first height (H.sub.1), while the second flap sheet **108***b* is illustrated as having a second width (W.sub.2) and a second height (H.sub.2). The first width (W.sub.1) is greater than the second width (W.sub.2) and the first height (H.sub.1) is greater than the second height (H.sub.2). As a result, as best illustrated in FIG. **5***b*, the first flap sheet **108***a* defines an overhang on each side of the flap assembly **108** having a first distance (D.sub.1). The first distance (D.sub.1) being the first width (W.sub.1) minus the second width (W.sub.2) and then divided by two. Similarly, as best illustrated in FIG. **5***c*, the first flap sheet **108***a* defines an overhang on the free edge **212***b* having a second distance (D.sub.2). The second distance (D.sub.2) being the first width (W.sub.1) minus the second width (W.sub.2). In some examples, the first distance (D.sub.1) is substantially equal to the second distance (D.sub.2). Notably, the first flap sheet **108***a* and the second flap sheet **108***b* are aligned at the attachment edge **212***a* (i.e., there is no overlap distance) to enable the attachment edge **212***a* of the valve flap assembly **108** to remain sandwiched between the bearing surface **302** and an inner surface of the bracket **304**. Likewise, the flap openings **308** formed along the attachment edge **212***a* of each of the first flap sheet **108***a* and the second flap sheet **108***b* align to permit attachment to the fastening projections **210**.

(37) While the present method and/or system has been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made

and equivalents may be substituted without departing from the scope of the present method and/or system. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. For example, block and/or components of disclosed examples may be combined, divided, re-arranged, and/or otherwise modified. Therefore, the present method and/or system are not limited to the particular implementations disclosed. Instead, the present method and/or system will include all implementations falling within the scope of the appended claims, both literally and under the doctrine of equivalents.

Claims

1. A pressure vent valve for ventilation of an interior of a vehicle, the pressure vent valve comprising: a frame defining a vent opening, at least one sealing surface, and a plurality of fastening projections, wherein the at least one sealing surface and the plurality of fastening projections are positioned adjacent the vent opening; and a valve flap assembly mounted on the frame at a bearing surface via the plurality of fastening projections, wherein the valve flap assembly comprises a first flap sheet stacked upon a second flap sheet, wherein, in an idle position, the valve flap assembly abuts the sealing surface and is configured to block a flow of air through the vent opening, and wherein, when a pressure increases in the interior of the vehicle, the valve flap assembly assumes an open position where the valve flap assembly is lifted from the sealing surface to allow the flow of air through the vent opening.
2. The pressure vent valve of claim 1, wherein the valve flap assembly has a thickness between about 0.75 mm and 1.25 mm.
3. The pressure vent valve of claim 2, wherein the valve flap assembly has a thickness of about 1.0 mm.
4. The pressure vent valve of claim 2, wherein each of the first flap sheet and the second flap sheet has a thickness of about 0.5 mm.
5. The pressure vent valve of claim 1, further comprising a second valve flap assembly mounted on the frame and configure to block a flow of air through a second vent opening formed in the frame.
6. The pressure vent valve of claim 1, wherein the frame is fabricated from a rigid material and the valve flap assembly is fabricated from a flexible material.
7. The pressure vent valve of claim 6, wherein the flexible material is a closed pore, foamed material.
8. The pressure vent valve of claim 6, wherein the flexible material is a plastic material or a rubber material.
9. The pressure vent valve of claim 1, wherein the frame further defines a stop surface and the valve flap assembly is secured to the frame such that the valve flap assembly assumes the open position by bending at a location that is spaced from the plurality of fastening projections.
10. The pressure vent valve of claim 9, wherein the stop surface extends over a portion of the valve flap assembly such that, upon assuming the open position, the portion covered by the stop surface is undeflected.
11. The pressure vent valve of claim 1, wherein the valve flap assembly bends in a curved manner when in the open position.
12. A pressure vent valve for ventilation of an interior of a vehicle, the pressure vent valve comprising: a frame defining a plurality of vent openings, wherein a sealing surface is positioned adjacent each of the plurality of vent openings; and a plurality of valve flap assemblies mounted on the frame, each of the plurality of valve flap assemblies comprising a first flap sheet stacked upon a second flap sheet, wherein one of the plurality of valve flap assemblies is coupled to the frame at each of the plurality of vent openings and configured to alternate between an idle position and an open position, wherein, in the idle position, the one of the plurality of valve flap assemblies is

configured to block a flow of air through one of the plurality of vent openings, and wherein, in the open position, the one of the plurality of valve flap assemblies is lifted from the sealing surface to allow the flow of air through the one of the plurality of vent openings.
13. The pressure vent valve of claim 12, wherein at least one of the plurality of valve flap assemblies has a thickness between about 0.75 mm and 1.25 mm.
14. The pressure vent valve of claim 13, wherein each of the first flap sheet and the second flap sheet has a thickness of between about 0.375 mm and 0.75 mm.
15. The pressure vent valve of claim 12, wherein the frame is fabricated from a rigid material and the plurality of valve flap assemblies is fabricated from a flexible material.
16. The pressure vent valve of claim 15, wherein the flexible material is a closed pore, foamed material.
17. The pressure vent valve of claim 15, wherein the flexible material is a plastic material or a rubber material.
18. The pressure vent valve of claim 12, wherein the frame further defines a stop surface and at least one of the plurality of valve flap assemblies is secured to the frame such that the at least one of the plurality of valve flap assemblies assumes the open position by bending at a location that is spaced from a plurality of fastening projections.
19. The pressure vent valve of claim 18, wherein the stop surface extends over a portion of the at least one of the plurality of valve flap assemblies such that, upon assuming the open position, the portion covered by the stop surface is undeflected.
20. The pressure vent valve of claim 19, wherein the plurality of valve flap assemblies bend in a curved manner when in the open position.